

Class 12: HW Population Analysis Pt.2

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Section 1. Proportion of G/G in a population

Downloaded a CSV file from ensemble->https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=39895595;v=rs8067378;vdb=variation;vf=959672880#_

```
MXL<- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(MXL)
```

	Sample..	Male.	Female.	Unknown.	Genotype..	forward.	strand.	Population.s.	Father
1					NA19648	(F)		A A ALL, AMR, MXL	-
2					NA19649	(M)		G G ALL, AMR, MXL	-
3					NA19651	(F)		A A ALL, AMR, MXL	-
4					NA19652	(M)		G G ALL, AMR, MXL	-
5					NA19654	(F)		G G ALL, AMR, MXL	-
6					NA19655	(M)		A G ALL, AMR, MXL	-
	Mother								
1	-								
2	-								
3	-								
4	-								
5	-								
6	-								

```
round(table(MXL$Genotype..forward.strand.)/nrow(MXL)*100,2)
```

A A	A G	G A	G G
34.38	32.81	18.75	14.06

Lets look at the GBR dataset

```
GBR<-read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

```
round(table(GBR$Genotype..forward.strand.)/nrow(GBR)*100,2)
```

A A	A G	G A	G G
25.27	18.68	26.37	29.67

This variant (GG) that is associated with childhood asthma is more frequent in GBR populations than MXL populations.

Further dig.

Homework Problems

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
pop<- read.table("rs8067378_ENSG00000172057.6.txt")  
nrow(pop)
```

```
[1] 462
```

```
table(pop$geno)
```

A/A	A/G	G/G
108	233	121

```
?median()
```

```
head(pop)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

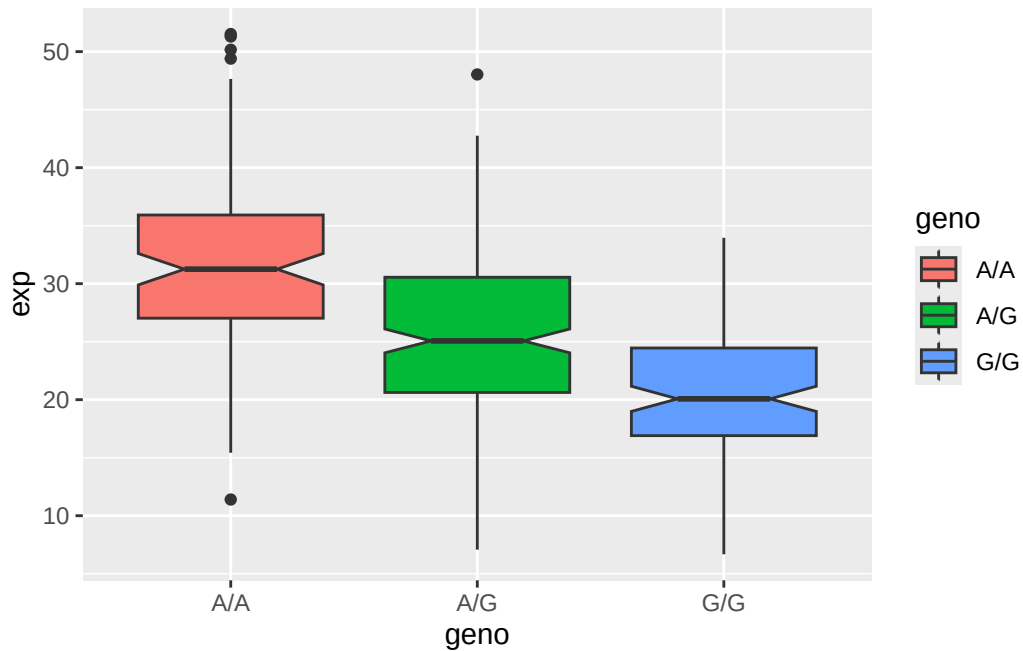
```
pop|>
  arrange(geno)|>
  group_by(geno)|>
  summarize(MedianExp<-median(exp))
```

```
# A tibble: 3 x 2
  geno `MedianExp <- median(exp)`
  <chr>                <dbl>
1 A/A                  31.2
2 A/G                  25.1
3 G/G                  20.1
```

Sample size= 462 AA: 108 (median expression:31.2), AG:233(median expression:25.1), GG:121(median expression: 20.1)

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
ggplot(pop)+
  aes(x=geno, y=exp, fill=geno)+
  geom_boxplot(notch=TRUE)
```



You can infer using the above box plot, that genotyping has a correlation with expression levels. With GG showing lower expression levels than AA. Yes these SNP variation do indeed seem to play a role in ORMDL3 expression, so depending on the variation this will either have an increase or decrease in expression levels.